

Exploring the evolutionary characteristics of social media data in metro emergencies: A case study of Zhengzhou Metro flood

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ABSTRACT

With the development of urban transportation, metros have become an important means of travel for residents. However, casualty and economic loss might occur in metro systems due to various emergencies. Social media has gradually become the main way to express people's needs, which provides a new analysis perspective for risk management in metros. This study takes the Zhengzhou Metro flood as an example and collects relevant social media data. Then, the analysis method of social media data evolution characteristics in metro emergencies is proposed. Finally, the evolution characteristics of social media data are analyzed from three aspects: spatiotemporal distribution, emotional distribution, and hot topics classification. The results show that: The temporal distribution of social media data is affected by the emergency process and official media; the spatial distribution of social media data reflects the distribution of stations affected by emergency and temporary shelters; timely and appropriate official media reports are conducive to guiding public emotions toward positive; and the key hot topics can be divided into disaster environment (DE), disaster impact (DI), disaster carrier (DC), emergency management (EM), positive comment (PC), and negative comment (NC). The proposed method can provide support for public opinion analysis and risk management in metro emergencies.

1 Introduction

In recent years, metro transport, as an important component of public transport in China's large cities, has been promoted in increasing numbers of cities. By the end of 2021, 50 cities in mainland China had developed metro systems with a total length of over 7000 km [1]. Meanwhile, China is one of the countries seriously affected by natural disasters. Especially in coastal areas, the safe operation of the metro system is vulnerable to typhoons and extreme rainstorms. Thus, severe casualty and economic loss might occur in the

operation and management of the metro system due to the coupling of various emergencies and sudden large passenger flow [2]. In the process of emergency response, distress messages in social media provide key information for the emergency rescue of the decision-making department. In addition, the emotional tendency of residents is also a direct expression of whether they are satisfied with the measures taken by the decision-making department [3]. Therefore, real-time monitoring and analysis of social media data can provide support for metro emergency response.

With the popularization of internet technology, social

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media has become an important means for people to express their opinions. Therefore, social media provides a new approach and challenge for the field of public safety and emergency management (EM). Nowadays, social media data have been widely used in the study of emergency management. As one of the most active social media, Twitter has become the main research object of scholars around the world. Carley et al. [4] assessed the value of Twitter for disaster management and pointed out that opinion leaders played an important role in disaster management, especially in the planning and early warning stages. Landwehr et al. [5] proposed the necessity for planning, warning, and response to tsunami disasters through Twitter. In China, the social media used by researchers is mainly Sina-Weibo (usually called Chinese Twitter). Shan et al. [6] proposed a disaster loss assessment model through Sina-Weibo data, which was used for the case studies of the Tianjin explosion accident and typhoon Nepartak. Wu et al. [7] discussed the change in public awareness of disaster risk in the process of emergency response and extracted disaster information from Sina-Weibo texts. Wang et al. [8] proved that Sina-Weibo is a good data source for the analysis of disaster management and urban sustainable development, and its distribution pattern is related to emergency characteristics. However, in emergency management, most of the research on social media focuses on urban natural disasters, which rarely focuses on the emergency evolution in the metro system. What is more, most of the existing studies related to metro social media data are dedicated to the analysis of the spatial and temporal characteristics of the data in daily operations. For example, Li et al. [9] took the Nanjing metro as an example to summarize the spatial and temporal distribution of social media data and extract public opinions from social media, which proposed improvement measures for metro operation. Luo et al. [10] took Shenzhen Metro as an example to explore the spatio-temporal dynamics of perceptions of metro services based on social media data. The feasibility of social media data in metro management research has been confirmed in existing studies. However, the research on metro social media has not been refined into the response process of a specific emergency. Therefore, it is necessary to explore the evolution characteristics of social media data in combination with actual metro emergencies.

At present, the research on the evolution characteristics of social media data mainly focuses on the spatio-temporal distribution characteristics, emotional distribution characteristics and hot topic classification. In the analysis of temporal distribution characteristics, He [11] combined grey system theory and fuzzy neural network to propose a prediction algorithm to predicate the number of social media data. Zhao et al. [12] proved that the time sequence change

of social media data of emergencies was consistent with the life cycle of disasters by analyzing the temporal distribution of social media data. In the analysis of spatial distribution characteristics, the main way to mine the locations of social media users is to resolve the identification of position (IP) address or social media text. Tang et al. [13] obtained the location information of social media data by combining the user registration positions, Sina-Weibo locations based on IP address, and the spatial information mentioned in the text, so as to study the risk perception in typhoon Lekima. Peng et al. [14] identified the community address of pneumonia patients from the Sina-Weibo text to analyze the spatial characteristics of the spread of COVID-19 in Wuhan. Xing et al. [15] determined the spatial location of social media data by structuring a place name database of the research area and using string matching in the analysis of the Jiuzhaigou earthquake. However, the spatial information obtained through IP addresses is generally accurate to the administrative units at the county level. In metro emergency management, the location information that needs to be controlled should be accurate to a specific station or an interval between stations. Therefore, the location information obtained through IP address cannot fully meet the needs of metro emergency management, which brings challenges to the analysis of spatial distribution characteristics of social media data in metro emergencies.

For emotional characteristics analysis, researchers mainly show solicitude for the emotional changes of the public before and after the emergency. Garske et al. [16] and Gruebner et al. [17] studied the distribution and change of negative emotions in New York City before, during and after superstorm Sandy, and found that the negative emotions of the public were correlated with the evolution of disasters. In analysis methods, scholars divided the emotional characteristics of social media data into positive, negative, and neutral levels by calculating the emotional value from social media texts. The methods of calculating emotional value mainly include the Bayesian algorithm [18, 19] and machine learning [20–22]. In the process of emergency response, negative emotions are usually concerned by researchers and managers in metro safety. Paying attention to the negative emotions among the affected groups is beneficial to detect and avoid public fear and crisis [23]. Therefore, in order to guide people's emotions to be positive in the emergency response process, it is a major work for metro public opinion management to identify the periods with low emotions on social media and analyze the reasons for the reduction of emotional value.

In the analysis of hot topic classification, some researchers focus on the improvement and application of clustering methods in social media hot topic mining. Xu et al. [24] used the latent Dirichlet allocation (LDA)

model to extract topic information from news texts and explained that the LDA model had higher accuracy than the *K*-means clustering method. Li et al. [25] analyzed and compared the application of word co-occurrence networks and improved single-pass algorithm in social media topic discovery. However, the topic classification results achieved by clustering methods are affected by the number of clusters. Besides, there are great differences and uncertainties in the hot topic classification of different events by clustering methods. Furthermore, too many topic classifications may not be conducive to public opinion management in the field of public safety. Therefore, in metro emergency management, determining how to classify topics of public concern reasonably and discussing the evolution of these topics in the emergency process are still difficult points in the current work.

This study aims to explore the evolutionary characteristics of social media data in metro emergencies. The metro emergency is taken as the research object to collect social media data from Sina-Weibo. Then, a method to analyze the evolutionary characteristics of social media data is proposed, which is applicable to the safe operation of metros in case of

emergencies. After that, the Zhengzhou Metro flood is taken as an example to analyze the evolutionary characteristics of social media data from three aspects: spatio-temporal distribution characteristics, emotional distribution characteristics, and hot topics classification characteristics. The analysis of the evolutionary characteristics of social media data can provide guidance for the safe operation and public opinion guidance of metros.

2 Methodology

In this study, the social media data in Sina-Weibo are collected by web crawler technology, and the emergency disposal processes of metro emergencies are obtained by the accident investigation report issued by the government. Then, natural language processing (NLP) technology is used to manipulate the texts of social media data. Finally, the evolutionary characteristics of social media data are analyzed based on the results of NLP. The technical methods used in this paper are divided into four levels: data, technology, application, and target, as shown in Fig. 1.

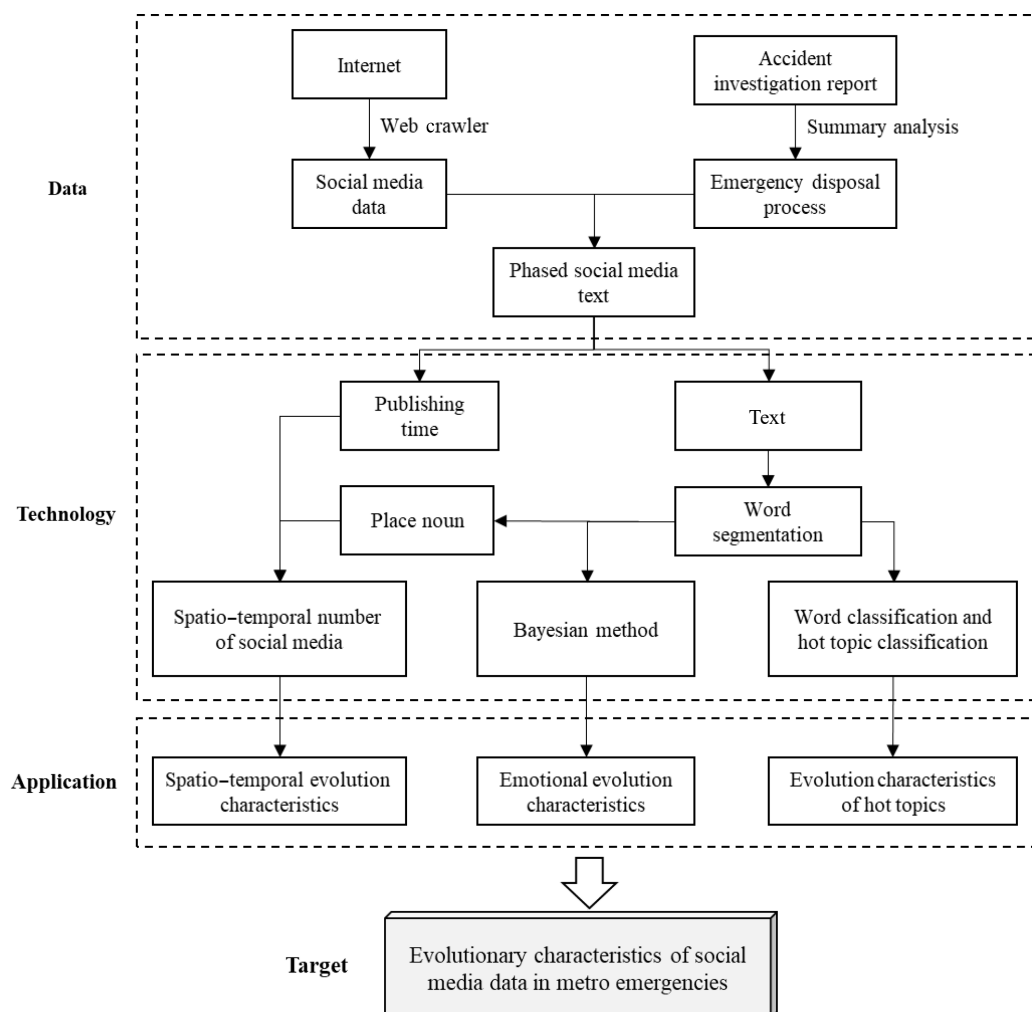


Fig. 1 4-level technical methods in the study.

2.1 Spatio-temporal evolution characteristics

2.1.1 Temporal evolution characteristics

To analyze the temporal evolution characteristics of social media data in metro emergencies, the publishing time of each social media data must be obtained. Web Spider is a computer technology that automatically and efficiently browses the internet and captures the required data. So, massive Sina-Weibo data published on the network during emergencies can be obtained by web crawler technology. The data collected in this paper include text, publishing time, author, forwarding number, and likes number, among which text and publishing time are the focus of this paper. the i^{th} social media data (Y_i) are shown in Eq. (1)

$$Y_i = (C_i, t_i), 1 \leq i \leq N \quad (1)$$

where C_i represents the text of the i^{th} social media data, t_i represents the publishing time of the i^{th} social media data, and N is the total number of social media data.

For a day, every 5 min is marked as a period, and 24 h can be divided into 288 periods. Then, the number of social media data of each period is counted as the 5-min social media popularity. The social media popularity (E_p) of p^{th} period is calculated as follows

$$E_p = \sum_{i=1}^N p(i) \quad (2)$$

$$p(i) = \begin{cases} 1, & t_i \in T_p \\ 0, & t_i \notin T_p \end{cases} \quad (3)$$

where $p(i)$ is a binary variable indicating whether the i^{th} social media data are published in T_p . T_p is marked as the time range of the p^{th} period, where p is the order number of the period, and $p = 1, 2, \dots, n, \dots, 288$. For example, when 08:00 is taken as the beginning of a day, T_1 is 08:00–08:05, T_2 is 08:05–08:10, and so on. So, the 5-min social media popularity can be obtained by the above method and the change curve of popularity can be drawn.

2.1.2 Spatial distribution characteristics

In this section, a method of the spatial distribution characteristic analysis of social media data for response to metro emergencies is proposed. To analyze the spatial distribution of social media data, the location of each social media data must be obtained. The geographic location information of cities and counties can be obtained through the IP address of social media data. However, this study needs to obtain the location information accurate to a specific metro line or station which cannot be obtained directly by the IP address of social media data. Therefore, a location information

recognition method based on text is proposed and used to identify metro lines or stations in social media data.

(1) Construct the metro line name corpus (LNC) and metro station name corpus (SNC).

We assume that there are l metro lines (marked as Line 1, Line 2, ..., Line l) and s metro stations (marked as $S_1, S_2, \dots, S_s$) in a city. Then, keyword lists are built for each metro line and station to construct metro LNC and metro SNC. An example of constructing LNC and SNC is shown in Table 1. If the words in the keyword list appear in the social media text, the location information of the social media data is the corresponding line or station.

(2) Initialize the spatial distribution of social media popularity.

E_L is marked as the social media popularity of metro line, while E_S is marked as the social media popularity of metro station. Initialize to $E_L = 0$ and $E_S = 0$, where $L \in \{\text{Line 1, Line 2, ..., Line } l\}$ and $S \in \{S_1, S_2, \dots, S_s\}$.

(3) Calculate the social media popularity of every metro line and station.

First, for each social media data, judge whether the text contains the keywords in LNC or SNC. Then, if the text contains the keywords in LNC or SNC, the social media popularity of the corresponding line or station increases by 1. The pseudo code for calculating the social media popularity of each line or station is shown in Table 2.

(4) Draw the popularity distribution of metro lines and stations in the metro map.

After obtaining the social media popularity of each metro line and station by the above process, geographic information system (GIS) technology is used to draw the popularity distribution of metro lines and stations in the metro map, so as to analyze the spatial distribution of social media data in metro emergencies.

2.2 Emotional evolution characteristics

Text emotion analysis (also known as opinion mining) is an important research branch in the field of NLP, which is a mining technology of text views and emotions through computer technology, and finally classifies and judges the emotional tendencies of text. In this study, a naive Bayes algorithm is used to analyze the emotion of social media data in metro emergencies. The basic principle of the algorithm is as follows.

Table 1 Constructions of LNC and SNC

Type	Name	Keyword list
LNC	Line 1	Line 1, line one
	Line 2	Line 2, line two
	...	...
SNC	SKL station	Shakoulu, SKL station
	HTS station	Haitansi, HTS station
	...	...

Table 2 Calculation method for social media popularity

Algorithm
For i from 1 to N :
Obtain C_i
For L in {Line 1, Line 2, ..., Line l }:
If the keyword about L is in C_i :
$E_L += 1$
End
For S in $\{S_1, S_2, \dots, S_s\}$:
If the keyword about S is in C_i :
$E_s += 1$
End
End

Suppose D is a given emotional text library, and d is the text in D . Firstly, we need to eliminate unnecessary adverbs, auxiliary words, pause words, and other meaningless words in the text. Then, $W(w_1, w_2, \dots, w_n)$ is marked as the feature item of the texts to be classified, and $C(c_1, c_2, \dots, c_n)$ is the classification set. Finally, the probability that the text emotion belongs to each classification (marked as $P(c_m|W)$) is calculated, and the c_m that maximizes $P(c_m|W)$ is taken as the classification of the text. There are three emotional classifications in this study: positive emotion (c_1), negative emotion (c_2), and neutral emotion (c_3). $P(c_m|W)$ is calculated as follows [26]

$$P(c_i|w_1, w_2, \dots, w_n) = \frac{P(w_1, w_2, \dots, w_n|c_i)P(c_i)}{P(w_1, w_2, \dots, w_n)}, \quad (4)$$

$$i = 1, 2, 3$$

where $P(w_1, w_2, \dots, w_n)$ is the probability of the emotional feature word $w_1, w_2, \dots, w_n$ appearing in D , and $P(w_1, w_2, \dots, w_n)$ is a constant when D is fixed; $P(c_i)$ is prior probability, and $P(c_i) = \frac{N_{c_i}}{N_{ts}}$, where N_{ts} is the number of training samples, and N_{c_i} is the number of training samples that belong to classification c_i . According to the naive Bayes algorithm, each feature item is independent of each other [26], hence

$$P(w_1, w_2, \dots, w_n|c_i) = \prod_{j=1}^n P(w_j|c_i), \quad i = 1, 2, 3 \quad (5)$$

For a text d to be classified, calculate $P(d|c_i)P(c_i)$ of each classification c_i . The classification that maximizes the probability is the classification of the text. The classification of the text d ($c(d)$) is determined as follow

$$c(d) = \operatorname{argmax} P(c_i) \prod_{j=1}^n P(w_j|c_i), \quad i = 1, 2, 3 \quad (6)$$

The emotional analysis text library used in this study is the open source text library included in SnowNLP that is a Python class library for Chinese NLP.

2.3 Evolution characteristics of hot topics

Word frequency analysis counts and analyzes the number of important words in text, which is the basis for mining hot topics in social media data. Its basic principle is to determine the hot topics and their change trends by the change of word frequency. Traditional word frequency analysis includes the following two steps.

(1) Word segmentation.

Word segmentation is the problem of dividing a string of written language into its component words [27]. Chinese word segmentation refers to dividing a Chinese character sequence into individual words. The rules of word segmentation in Chinese text are different from those in English text, and we cannot use spaces to divide words like English word segmentation. Therefore, scholars usually use the Jieba library to segment Chinese text [28].

(2) Stop word filtering and word frequency count.

Stop words are an important tool in the field of NLP, which are usually used to improve the quality of text features or reduce the dimensions of text features [29]. In the social media text of metro emergencies, punctuation marks, mood words, and other meaningless characters are interference characters that should be eliminated. After stop word filtering, the number of occurrences of keywords is counted to get the high-frequency hot words in the social media text.

However, the traditional word frequency analysis method can be further improved. One reason is that the importance of keywords in social media texts cannot be fully characterized only by the number of occurrences. What is more, multiple words may represent the same topic, so these words can be integrated and classified. At present, there are fewer studies on the hot topic classification of social media data in metro emergencies. This study classifies the keywords of social media data in metro emergencies based on existing research on term frequency-inverse document frequency (TF-IDF) [30], public security triangular framework [31], etc. On the basis of Chinese word segmentation, stop word filtering and word frequency count, the following steps are further carried out.

(1) Calculate TF-IDF.

TF-IDF is used to evaluate the importance of a word in the social media text corpus, and its calculation method is as follows

$$TF = TF(w) \times IDF(w) \quad (7)$$

$$TF(w) = \frac{n_w}{\sum_i n_i} \quad (8)$$

$$IDF(w) = \log \left(\frac{N}{1 + |\{j: w \in C_j\}|} \right) \quad (9)$$

where n_w is the number of occurrences of the word w in the text, $\sum_i n_i$ is the total number of occurrences of all words in the text, and $|\{j: w \in C_j\}|$ is the number of texts containing the word w .

(2) Reduce the dimension of text.

After calculating TF-IDF, each Y_i can be represented by multiple keywords $w_1, w_2, \dots, w_m$ and their corresponding TF-IDF

$$Y_i = \{w_1: \text{TF-IDF}(w_1), w_2: \text{TF-IDF}(w_2), \dots, w_m: \text{TF-IDF}(w_m)\} \quad (10)$$

In Eq. (10), $\text{TF-IDF}(w_i)$ is the TF-IDF of the word w_i , and m is the dimension of the social media data Y_i .

If the social media text is too long, the dimension of the feature vector will be very large. A high dimension makes clustering difficult and increases computational complexity. On the other hand, a dimension that is too small may lead to data distortion and reduce the accuracy of the computed results. Based on existing research [32], in order to facilitate keyword classification, the maximum dimension used in this study is 50. If $m < 50$, all keywords and their TF-IDF are retained. If $m \geq 50$, the keywords are sorted according to TF-IDF, and the keywords with smaller TF-IDF are eliminated.

(3) Determine the topic classification of keywords.

The key problem of topic classification is determining the number and content of classification. The classification of topics should include all information on metro emergencies as much as possible. According to public security triangular framework [31], the contents of the public security include emergencies, disaster

carriers (DCs), and emergency management. Therefore, the social media data in metro emergencies can be classified into these three contents. In terms of emergencies, social media texts are divided into the descriptions of disaster environments (DEs) (Class 1) and the descriptions of disaster impacts (DIs) (Class 2). In terms of EM, the information involved includes the prevention, early warning, rescue, and recovery of metro emergencies, as well as the government's rescue measures and rescue progress (Class 3). The DCs include people, objects, and systems. In metro emergencies, the disaster-carrying system is the metro system, and the disaster-carrying objects are the component parts of the metro system, including metro stations, trains, and tunnels (Class 4). In addition, the people affected by metro emergencies include passengers and urban residents using the metro system, whose comments on social media are the key content of network public opinion research. Therefore, this study focuses on the comments of passengers and urban residents, including positive comments (PCs) (Class 5) and negative comments (NCs) (Class 6). The corresponding relationship between the public security triangular framework and hot topic classification is shown in Fig. 2. Therefore, the keywords in the social media text of metro emergencies are divided into 6 classifications: DE, DI, DC, EM, PC, and NC.

(4) Calculate the popularity of each topic classification.

After all keywords are classified into 6 classifications, the classification of each Y_i is determined. A social media text may contain the keywords of multiple

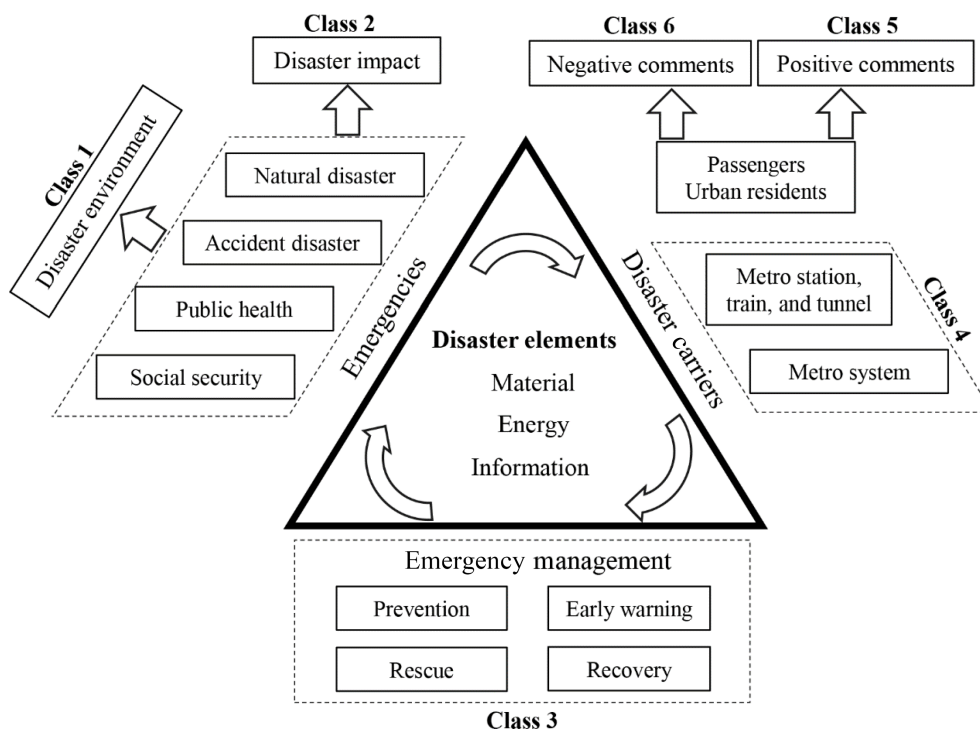


Fig. 2 Public security triangular framework and hot topic classification.

classifications, so this text belongs to multiple classifications at the same time and each classification has a weight. The weight of each classification for social media data Y_i is calculated by Eq. (11)

$$\text{Weight}(Y_i, \text{Class } j) = \frac{\sum_p \text{TF-IDF}(w_p)}{\sum_{k=1}^m \text{TF-IDF}(w_k)}, \quad (11)$$

$$w_p \in \text{Class } j, j = 1, 2, \dots, 6$$

where $\text{Weight}(Y_i, \text{Class } j)$ is the weight of classification Class j for social media data Y_i . The weight of all classifications meets the following Eq. (12)

$$\sum_{j=1}^6 \text{Weight}(Y_i, \text{Class } j) = 1 \quad (12)$$

Finally, the popularity and the proportion of each topic classification are calculated to analyze the evolution characteristics of hot topic classification. The popularity of Class j is calculated by Eq. (13)

$$E(\text{Class } j) = \sum_{i=1}^N \text{Weight}(Y_i, \text{Class } j) \quad (13)$$

3 Case study

3.1 Case overview

3.1.1 Zhengzhou Metro flood

Zhengzhou is a city in the north-central part of Henan Province, China. From 08:00 on July 17, 2021 to 08:00 on July 23, 2021, Zhengzhou had 5590 km² of accumulated rainfall of more than 400 mm and 2068 km² of accumulated rainfall of more than 600 mm. The heavy rain on July 20 in Zhengzhou flooded a large

area of the city. According to the report of China's meteorological authority, the hourly average rainfall and accumulated rainfall from 09:00 on July 19 to 13:00 on July 21 are collected, and the rainfall change curve is drawn as shown in Fig. 3. The horizontal axis in Fig. 3 represents the date and time. For example, "1909" means 09:00 on the 19th, "2010" means 10:00 on the 20th, and so on. The rainfall near the Wulongkou (WLK) stabling yard in Zhengzhou Metro started to increase from 13:00 on the 19th, and reached the maximum from 16:00 to 18:00 on the 20th, even exceeding 100 mm per hour.

On July 20, 2021, a serious flood occurred in the WLK stabling yard of Zhengzhou Metro Line 5 and its surrounding areas due to continuous extreme rainfall. At about 18:00, the flood burst the retaining wall of the inbound-outbound siding and entered the main line of Line 5, causing a train on Line 5 to be surrounded by floods. After the rescue work was completed, 14 passengers were killed. The map of Zhengzhou Metro and WLK stabling yard is shown in Fig. 4. The local schematic diagram of the WLK stabling yard, adjacent stations, and accident location is shown in Fig. 5.

3.1.2 Accident disposal process

According to the accident progress and response progress in the investigation report of the Zhengzhou Metro flood accident [33], the whole accident can be divided into 6 stages, as shown in Fig. 6.

(1) Stage 1: Continuous rainstorm.

On July 20, before the accident, a continuous rainstorm occurred in Zhengzhou. At 16:52, the flood entered the WLK stabling yard. Then, the water of the WLK stabling yard entered the main line of Line 5.

(2) Stage 2: Parking.

At 17:43, the train arrived at Haitansi (HTS) station

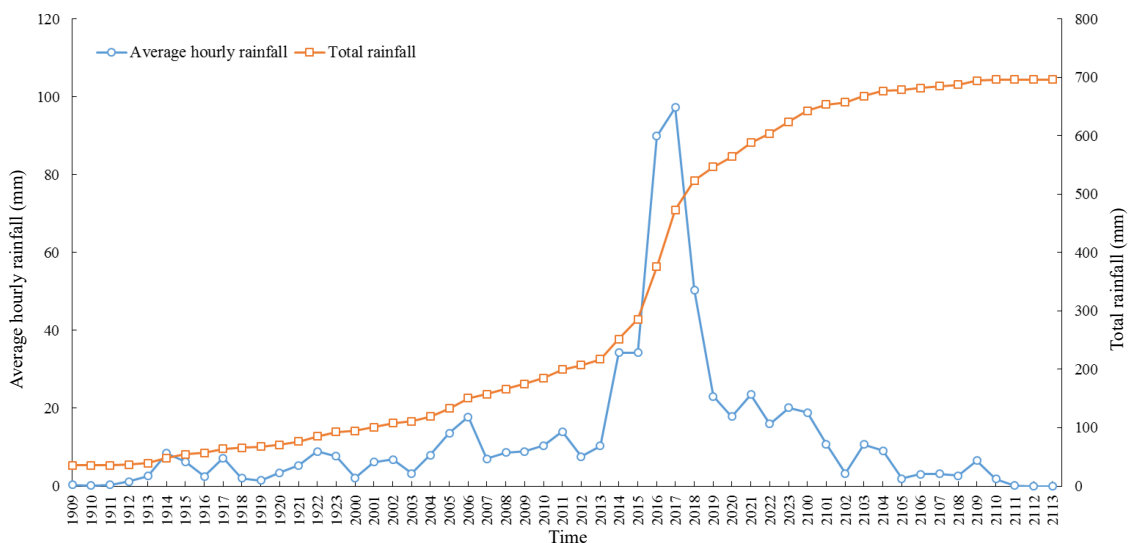


Fig. 3 Average hourly rainfall and total rainfall (near the WLK stabling yard).

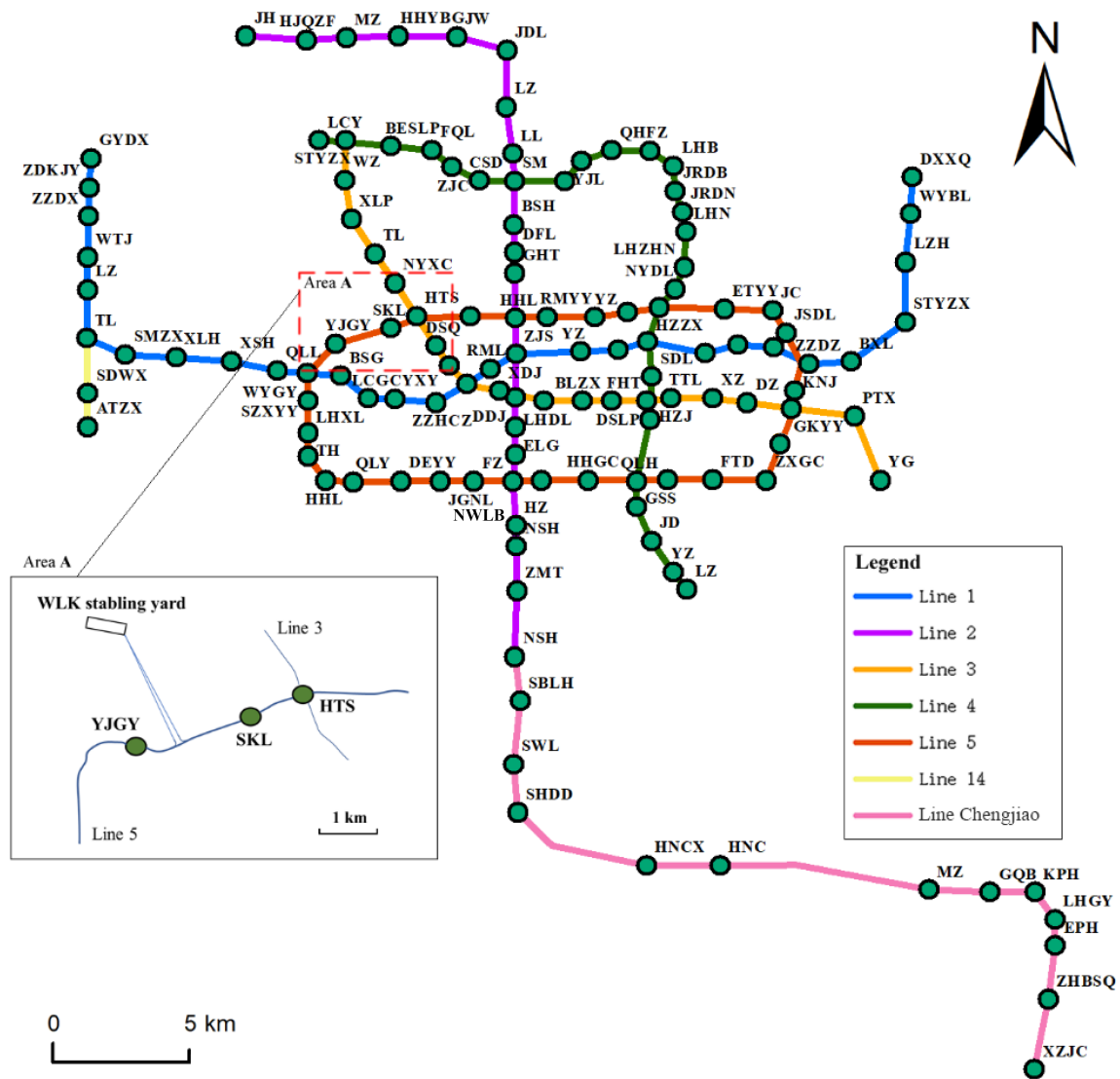


Fig. 4 Map of Zhengzhou Metro and WLK stabling yard.

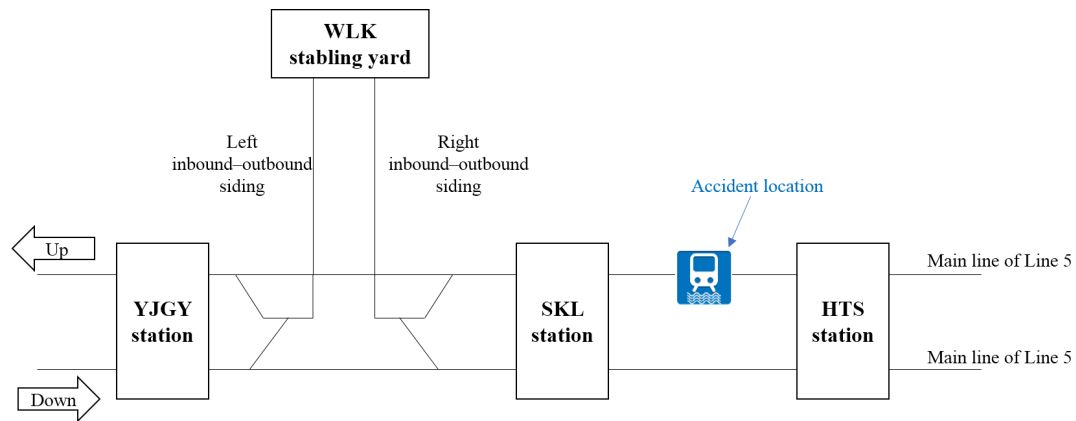


Fig. 5 Local schematic diagram of WLK stabling yard, adjacent stations, and accident location.

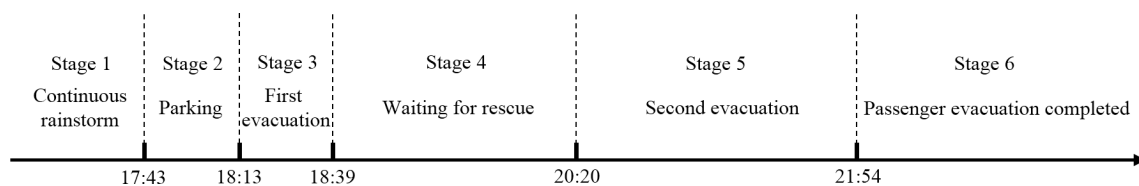


Fig. 6 Accident process of Zhengzhou Metro flood.

and was running toward Shakoulu (SKL) station. The monitoring equipment of the metro system gave an alarm because of the flood. So the train dispatcher set detaining in the up direction of HTS station of Line 5 (the first parking). However, as the train dispatcher did not find out the cause and did not know the seriousness of the situation, he informed the train to leave the HTS station at 17:46. Then, at 17:47, the flood crossed the rail surface and the train stopped (the second parking). At 17:58, the flood began to enter the carriage. At 18:07, the train was forced to stop at the tunnel from HTS station to SKL station due to a power failure.

(3) Stage 3: First evacuation.

At 18:13, the metro staff organized passengers to evacuate towards SKL station. Until 18:39, there were also more than 400 passengers who still did not complete evacuation. However, the flood had already overflowed the train door and evacuation was not possible.

(4) Stage 4: Waiting for rescue.

Between 18:39 and 20:20, the evacuation could not continue due to the flood. The passengers were informed to destroy the windows of the carriages to slow the rise of water level, strengthen ventilation, and wait for rescue.

(5) Stage 5: Second evacuation.

At 20:20, the passengers smashed the windows of the carriages with low water level, and the water level outside the carriages began to drop. At the same time, rescue workers arrived at the accident location. Then, the train doors were opened and evacuation continued. At 21:54, the last passenger in the carriage evacuated to the evacuation platform.

(6) Stage 6: Passenger evacuation completed.

When the last passenger in the carriage evacuated to the evacuation platform, the rescue workers continued to check whether there was any passenger left in the carriage. Finally, at 22:49, it was confirmed that all people had evacuated to the evacuation platform.

3.1.3 Cause analysis

Based on the accident investigation report of the Zhengzhou Metro flood, the causes of the accident are analyzed from three aspects: external environment, disaster-carrying buildings, and emergency management, as shown in Fig. 7. In terms of the external environment, a continuous rainstorm is only the inducing factor of the accident. In the case of multiple red warnings from China's meteorological authority, Zhengzhou Metro operators failed to launch emergency plans and take preventive measures in time, which become one of the important factors leading to the collapse of the water retaining wall in the WLK stabling yard. What is more, in terms of disaster-carrying buildings, poor drainage, poor enclosure quality, and the lack of maintenance are the important reasons for the collapse of the water retaining wall in the WLK stabling yard. In emergency management, the lack of great attention and chaotic management are the main causes of the accident. First, the train dispatcher took mistakes in commanding without finding out the cause and understanding the danger, which led to the forced stop of the metro train due to power failure. In addition, the metro operators did not launch the emergency plan in time, resulting in chaos in emergency management and

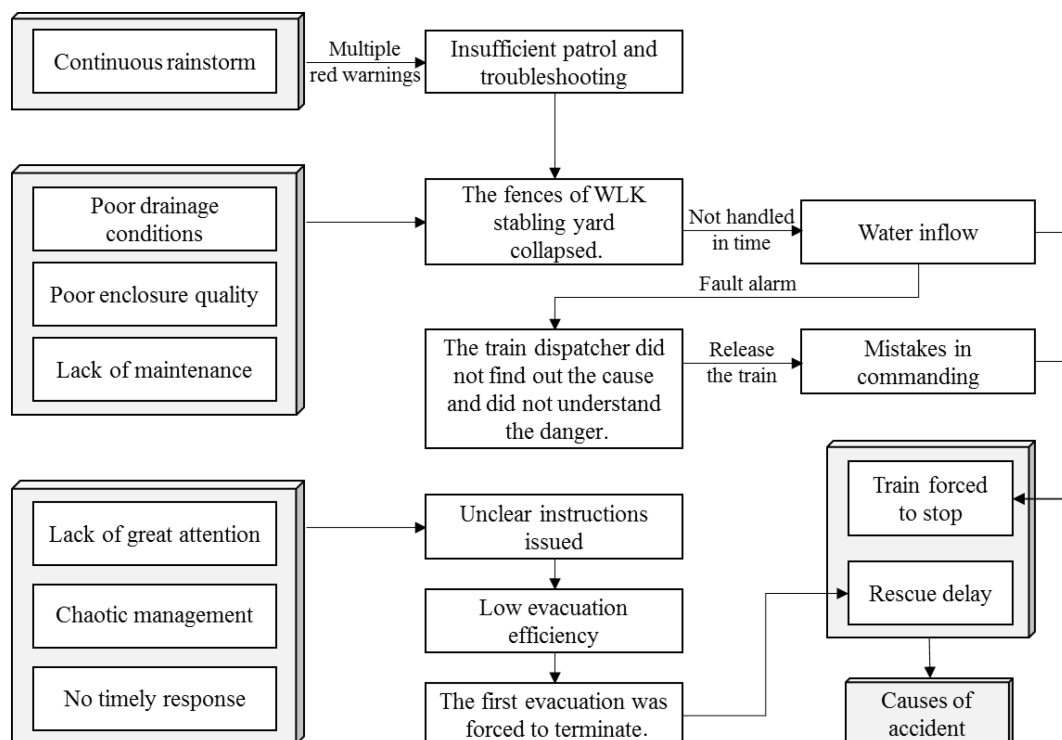


Fig. 7 Causes of Zhengzhou Metro flood accident.

delay in evacuation and rescue. According to Fig. 7, the lack of great attention is the important reason for mistakes in disaster warning, train dispatching, and emergency evacuation. If more importance could be attached to the impact of the rainstorm in the disposal process of emergency management, the loss of the accident would be reduced.

3.2 Spatio-temporal evolution characteristics of social media data

3.2.1 Temporal evolution characteristics

In this study, Sina-Weibo data from 08:00 on July 17 to 08:00 on July 23 are collected by web crawler technology. The collected Sina-Weibo data contain the keyword “metro” and any of the following words: “rainstorm”, “water flood”, “inundation”, “flood”, and “Zhengzhou”. A total of 42,168 social media data were collected. In the time range of collection, the number of social media data per day is shown in Fig. 8.

Before July 20, the impact of the rainstorm on the metro system did not acquire enough attention from the operation company of Zhengzhou Metro, indicating that Zhengzhou Metro still has problems in the warning stage of rainstorms. Before the accident, there are very little social media data about the metro and rainstorm, and the daily number of data did not exceed 700. The popularity of social media on July 19 is only 1.08% of that on July 20. It is worth noting that China’s meteorological authority has issued several red

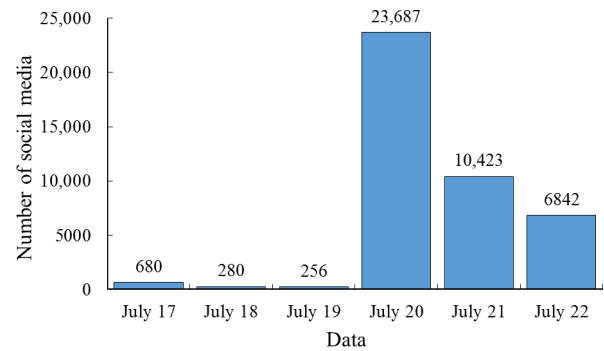


Fig. 8 Daily social media popularity of Zhengzhou Metro flood.

warnings since July 19. However, the metro system and rainstorm warning information did not receive attention on July 19. It can be seen that the public and metro operators believe that the rainstorm will not have a great impact on the metro system. In addition, the official media “@Zhengzhou Metro” published 5 messages on the 19th, as shown in Table 3, none of which mentioned the rainstorm and warnings. It shows that Zhengzhou Metro operators did not attach great importance to the rainstorm. The metro flood early warning mechanism in some inland cities in China is still imperfect. What is more, Zhengzhou Metro Line 5 has only 2 years of operation experience from May 2019 to the date of the accident. The employees of operation company of Zhengzhou Metro have a short length of service and lack experience in emergency response. Different from inland cities, coastal cities are vulnerable to typhoons

Table 3 Messages issued by official media on 19th and 20th

Symbol	Date	Time	Publishing department	Main content
—	19 th	08:31:25	Zhengzhou Metro	Morning peak passenger flow
—	19 th	10:05:54	Zhengzhou Metro	Tourism activities
—	19 th	13:06:53	Zhengzhou Metro	Cultural activities
—	19 th	15:21:35	Zhengzhou Metro	Safety knowledge
—	19 th	17:57:31	Zhengzhou Metro	Evening peak passenger flow
A	20 th	08:32:14	Zhengzhou Metro	Morning peak passenger flow
B	20 th	09:14:28	Zhengzhou Metro	Greeting
C	20 th	09:49:19	Zhengzhou Metro	Launch emergency plan
D	20 th	13:24:12	Zhengzhou Metro	Safety knowledge
E	20 th	16:30:05	Zhengzhou Metro	Running plan adjustment of Line 1 and Line 2
F	20 th	17:05:19	Zhengzhou Metro	Suspension of Line 4
G	20 th	17:09:45	Zhengzhou Metro	Running plan adjustment of Line 5
H	20 th	17:51:25	Zhengzhou Metro	Running plan adjustment of Line 5
I	20 th	18:42:17	Zhengzhou Metro	Suspension of whole metro system
J	20 th	18:48:23	Zhengzhou News Broadcasting	No damage in the rainstorm
K	20 th	19:23:57	Zhengzhou News Broadcasting	The evacuation of Line 5 is completed*
—	20 th	Between 20:30 to 21:00	Many news media such as The Paper News, Aikan News, and so on	The rescue of Zhengzhou Metro Line 5 is in progress
L	20 th	21:18:10	Zhengzhou News Broadcasting	The rescue is in progress
M	20 th	21:43:34	Zhengzhou News Broadcasting	The evacuation of Line 5 is completed

Note: *In fact, the evacuation was not completed at this time, and this news led to the public’s suspicion that the government was hiding.

and attach great importance to rainstorm warnings [34]. During the rainstorm warning, coastal cities such as Guangzhou and Shenzhen will take corresponding early warning measures, advise passengers to reduce unnecessary travel, and strengthen the inspection and management of metro operation [35]. On the analysis of social media popularity, Zhengzhou, as an inland city, pays less attention to rainstorms than coastal cities.

Then, the social media data on the day of the accident are discussed (from 08:00 on July 20 to 08:00 on July 21). A day is divided into 288 periods and every period is 5 min. The identifier of all periods are marked as id. The time period with id = 1 is from 08:00 to 08:05, and so on. The popularity change curve of social media data is drawn, corresponding to each stage of the Zhengzhou Metro flood accident, as shown in Fig. 9. The letters in the figure represent the messages identifier of the official media, as shown in Table 3.

The temporal evolution characteristics of social media popularity have obvious regularity in each stage of the Zhengzhou Metro flood accident. During the continuous rainstorm (Stage 1), the popularity of social media data is almost zero. As the flood gradually entered the main line, social media popularity began to rise. In the process of train parking, first evacuation and waiting for rescue (Stages 2–4, no external rescue participation), social media popularity steadily decreased. At this time, the emergency evacuation was mainly carried out by the trapped passengers and the metro staff. The reasons for the low popularity of social media from Stage 2 to Stage 4 are complex. First, the messages published by the official media mostly focus on the adjustment of the running plan rather than the train accident. Second, although the trapped passengers published some messages for help, the influence of these distress messages was low without widespread attention. Therefore, the accident disclosure and public opinion guidance of Zhengzhou Metro are insufficient and improper in emergencies. After the start of the second evacuation (Stage 5, with external rescue participation), social media popularity rose rapidly. The 5-min social media popularity rose from 100 to 590. In Stage 5, the news media gradually reported the rescue of the accident due to the participation of external rescue. Thus, the accident aroused wide public concern in a

short time. After the evacuation is completed (Stage 6), social media popularity gradually declined. The number of social media data did not stabilize until 2:00.

3.2.2 Spatial evolution characteristics

As of July 2021, Zhengzhou Metro has 7 lines and 131 stations, as shown in Fig. 4. Thus, the line name corpus and the SNC of Zhengzhou Metro are built as follows

$$\text{LNC} = \{\text{Line 1, Line 2, Line 3, Line 4, Line 5, Line 14, Line Chengjiao}\} \quad (14)$$

$$\text{SNC} = \{\text{SKL station, HTS station, HZZX station, ...}\} \quad (15)$$

LNC and SNC contain the names of all metro lines and all metro stations, respectively. The occurrence times of 7 lines in the social media texts are counted and the line distribution of social media data in Zhengzhou Metro flood, as shown in Table 4.

As the line where the accident occurred, Line 5 has the highest social media popularity. Of the total 42,168 social media data, 5423 are concerned about Zhengzhou Metro Line 5. The social media popularity of Line 4 is 2401, and it is the second highest social media popularity. Thus, the popularity of social media on metro lines could partly reflect the severity of the disaster of different lines. As the most seriously affected line in the flood with train shutdown and casualties, Line 5 has the highest social media popularity. Due to the water leakage at the exit of the DSLP station in Line 4, the flood entered the station hall, which affected the operation of Line 4. However, its disaster severity is less than that of Line 5, so the social media popularity of Line 4 is less than that of Line 5. The disaster severity of other lines is less than that of Line 5 and Line 4, so the social media popularity of other lines is not high. In addition, the small number of stations and remote locations are also one of the reasons for the low popularity of social media of other lines. For example, Line 14 only has three stations, and Line Chengjiao is far away from the city center, both of which have low social media popularity.

To explore the spatial evolution characteristics of

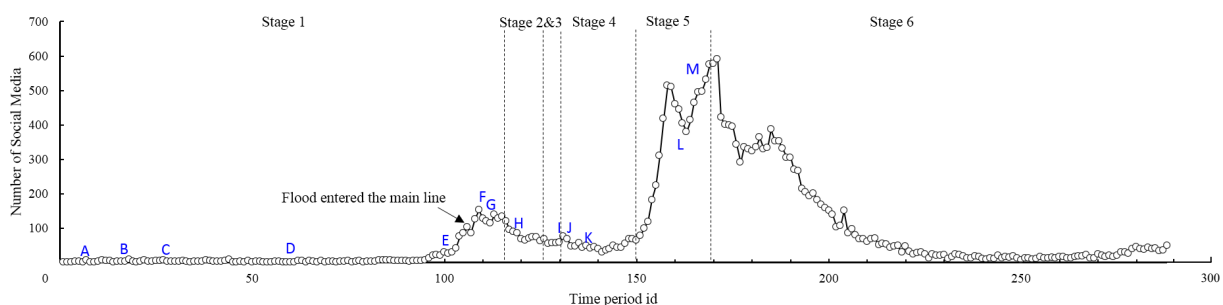


Fig. 9 5-min social media popularity of Zhengzhou Metro flood.

Table 4 Line distribution of social media in Zhengzhou Metro flood

Sort	Line name	Frequency	Number of stations	Fatality
1	Line 5	5423	32	14
2	Line 4	2401	27	0
3	Line 2	381	22	0
4	Line 1	321	30	0
5	Line 3	52	21	0
6	Line 14	18	3	0
7	Line Chengjiao	6	13	0

social media popularity in each line, the social media popularity of each line in the 6 stages is calculated, and the change curve is drawn, as shown in Fig. 10.

Before the second evacuation (Stages 1–4), the social media popularity of Line 4 is highest and gradually decreases. As the second evacuation begins (Stage 5), the social media data are mainly about Line 5 and is growing rapidly. After the evacuation is completed (Stage 6), the social media popularity of Line 5 are still the highest. Meanwhile, the social media popularity of Line 1 and Line 2, which has not received much attention, has increased, surpassing even that of Line 4. It shows that the public has paid more attention to the flood and started to care about the lines that have not yet had an accident after the accident of Line 5.

On the basis of the line distribution of social media data, the frequencies of 131 metro stations in 42,168 social media texts are calculated, as shown in Table 5. Then, the social media popularity of all stations is drawn on the map of Zhengzhou Metro. The spatial distribution of social media data is obtained, as shown in Fig. 11.

The metro stations with high social media popularity are those that are greatly affected by the disaster or have temporary shelters around them. Among 131 stations of Zhengzhou Metro, SKL station and HTS station have the highest social media popularity, because the SKL–HTS interval is the location where the train is forced to stop. While waiting for rescue, the

trapped passengers publish messages for help at the SKL–HTS interval. In addition, as an important transport hub with a large number of daily passengers, ZZDZ station also has a high social media popularity. On July 20, the first floor of ZZDZ station was flooded. Passengers who originally planned to travel through ZZDZ station published plenty of messages on social media, resulting in the high social media popularity of ZZDZ station. What is more, metro stations such as HZZX, ZZDX, NWLB, and NSH also have high social media popularity. The reason is that social personages provide free accommodation for the people who cannot return home near these stations and send messages on social media. Therefore, mapping the spatial distribution of social media data is helpful to find temporary shelters or the stations that are seriously affected by disasters.

3.3 Emotional evolution characteristics of social media data

The change of social media emotional value in each stage of the Zhengzhou Metro flood accident is calculated by the naive Bayes algorithm. In this study, the social media data from 08:00 on July 20 to 08:00 on July 21 are mainly discussed. As given in Section 3.2, a day is divided into 288 periods, and every period is 5 min. Firstly, in every 5 min, the emotional value of each social media text is calculated by the naive Bayes algorithm. Then, the average emotional value of all texts is calculated as the emotional value of the time period. Finally, the change curve of emotional value is drawn to analyze the emotional evolution characteristics of social media data in metro emergencies, as shown in Fig. 12. The letters in the figure represent the messages identifier of the official media (as shown in Table 3).

Emotional values reflect the emotional tendency of social media texts. By calculating the emotional values of all texts at a given moment, the public's emotional tendency towards a disaster event can be determined. Similar to the temporal evolution characteristics analysis mentioned earlier, this section analyzes the emotional values of the 6 stages of the Zhengzhou Metro flood.

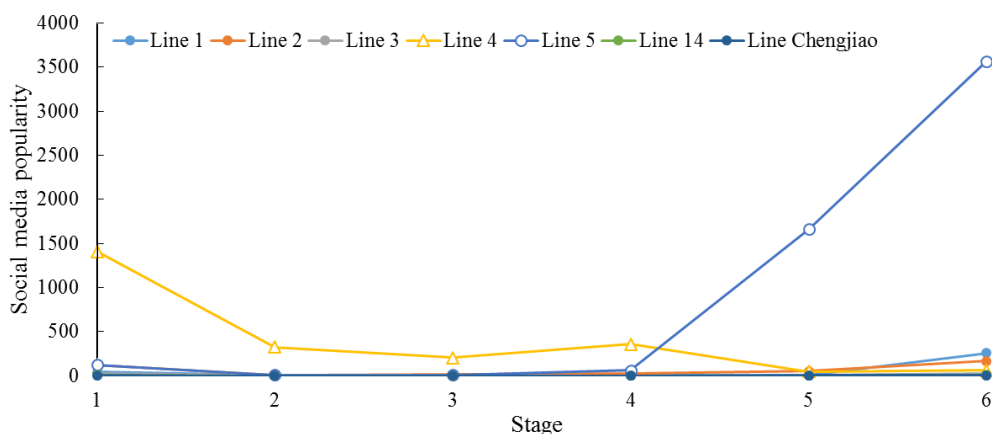
**Fig. 10** Evolutionary characteristics of social media line distribution in Zhengzhou Metro flood.

Table 5 Frequencies of stations in social media texts in Zhengzhou Metro flood

Sort	Station name	Line	Total frequency
1	SKL	5	319
2	HTS	3, 5	306
3	HZZX	1, 4	174
4	NWLB	2, 5	157
5	ZZDZ	1, 5	152
6	ZZDX	1	151
7	NSH	2	110
8	HHNL	1	90
9	DSLPL	3, 4	88
10	YZ	5	88
11	WTJ	1	81
12	SDL	4	81
13	BXL	1	72
14	DFL	2	68
15	ELG	2	68
16	JSDL	5	66
17	KNJ	5	64
18	XDJ	3	57
19	MHL	1	56
20	LZH	1	52
21	BSH	2	52
—	Other stations	—	<50

Since this study mainly focuses on the emotional changes of all people towards the disaster event, the time distribution of emotional values is mainly considered. Furthermore, as seen from Fig. 11, since the number of social media data at most stations is below 50, calculating emotional values may cause significant fluctuations. Therefore, analyzing the spatial distribution of emotional values does not have much practical significance in this case.

Before 16:00, the number of social media data every 5 min is less than 10. In this situation, the calculated emotional value is unstable and unrepresentative, which cannot express the real feeling of the public. Therefore, when the time period id ≤ 96 , the emotion value is set to null. Firstly, the naive Bayes algorithm is used to calculate the emotional value of social media texts which is a decimal between 0 and 1. The higher the emotional value, the more positive the emotion of the text. At present, most studies divide emotional value into three categories, and the texts with emotional value around 0.5 are classified as neutral [36]. Therefore, based on existing research, this study divides all emotional values into three categories: positive, neutral, and negative emotions. The emotional value of negative emotion is less than 0.4, the emotional value of positive emotion is more than 0.6, and the emotional value of neutral emotion is between 0.4 and 0.6.

During the continuous rainstorm (Stage 1), the public emotion changed from neutral emotion to negative

emotion. During the period from 16:00 to 17:40 (time period id is from 97 to 106), most people did not expect that the rainstorm would cause huge losses, and the emotion is mainly neutral. After 17:40, with the gradual increase of the rainstorm, people gradually realized the seriousness of the disaster. People began to worry about the development of the disaster, so their emotions changed from neutral to negative. During the train stop and the first evacuation (Stages 2 and 3), the public's emotion is still dominated by negative emotions. As more and more metro stations, shopping malls and buildings are flooded, the public published messages to express their concern. Meanwhile, in this stage, less official media reported the development and rescue of the accident. Due to the lack of correct media publicity and public opinion guidance, the characteristics of the public's emotions had been dominated by negative emotions. While waiting for rescue (Stage 4), the public's emotion is mainly neutral and has a gradual upward trend. The official media "@Zhengzhou Metro" published a message on social media at 18:42 and said that the whole network of Zhengzhou Metro has suspended operation services, please pay attention to safety. The official media "@Zhengzhou News Broadcast" published a message on social media at 18:48 and said that there are no casualties. The statements of some official media eased the public's concerns, so the emotional value was mainly neutral during this period. At this time, the train involved in the accident is waiting for rescue. Zhengzhou Metro has not announced the train accident of Line 5. Social media data still focused on the overall situation of the rainstorm, rather than the train accident on Line 5. During the second evacuation (Stage 5), the emotional characteristics of social media data are mainly neutral, but there is a declining tendency. According to the discussion in Section 3.2, at this stage, the social media popularity of Zhengzhou Metro Line 5 improves significantly. However, with the increase in social media popularity, emotional value has declined. The reasons are discussed as follows. The zero reporting system is an important system in China's metro operation, which stipulates that the situation of the occurred incident should be reported within a specified time even if no loss has been discovered. From Stage 2 to Stage 4, operation company of Zhengzhou Metro did not implement the zero reporting system. Facing the sudden surge of rescue messages, the public suspected the underreporting of the government. In addition, in this stage, most of the news media only reported the serious situation of the accident without emotional guidance at the same time. There are still deficiencies in the guidance of public opinion by operation company of Zhengzhou Metro and Zhengzhou news media. After the evacuation (Stage 6), the emotional characteristics of social media data changed from neutral to positive, and

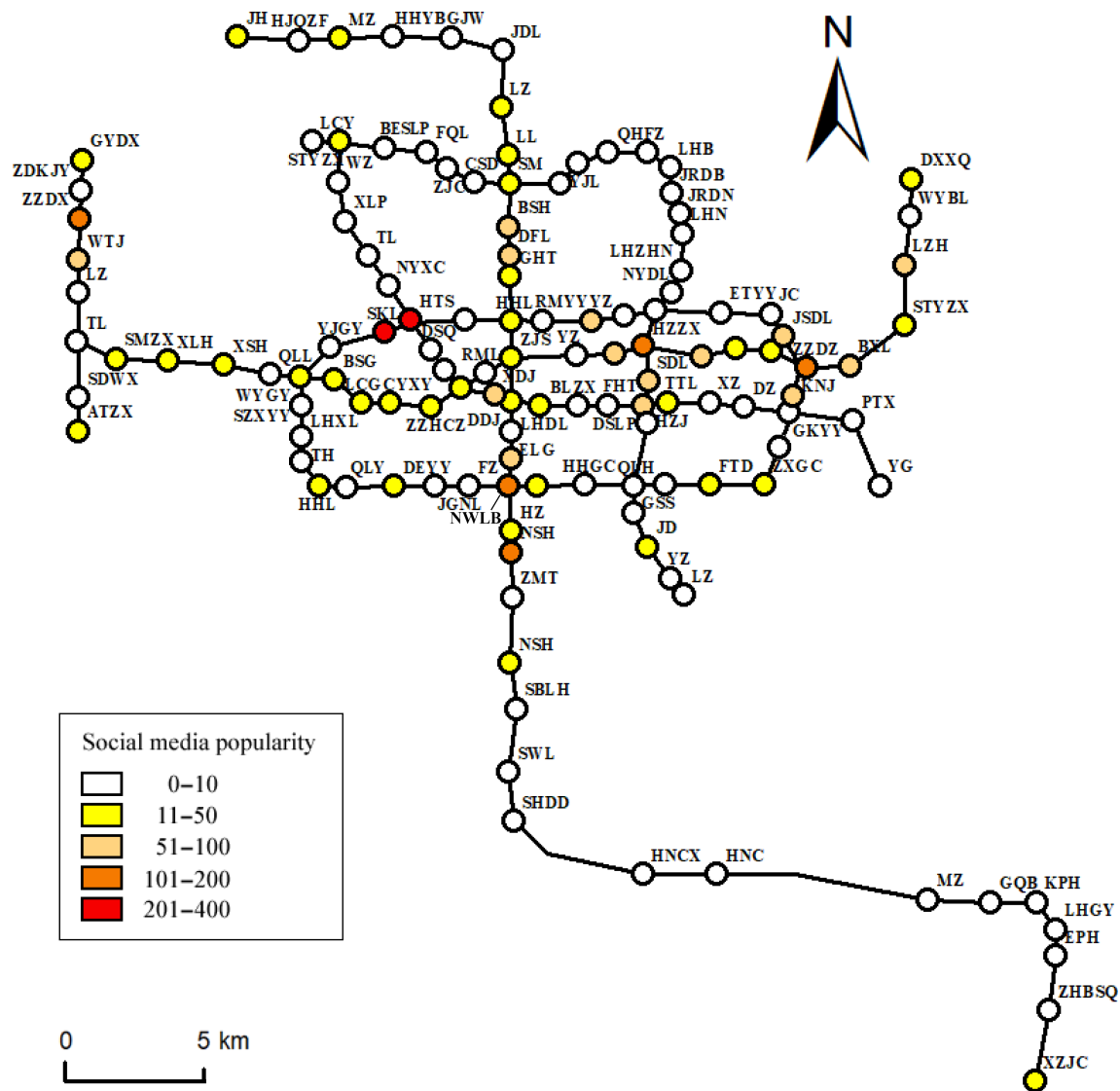


Fig. 11 Spatial distribution of social media data in Zhengzhou Metro flood.

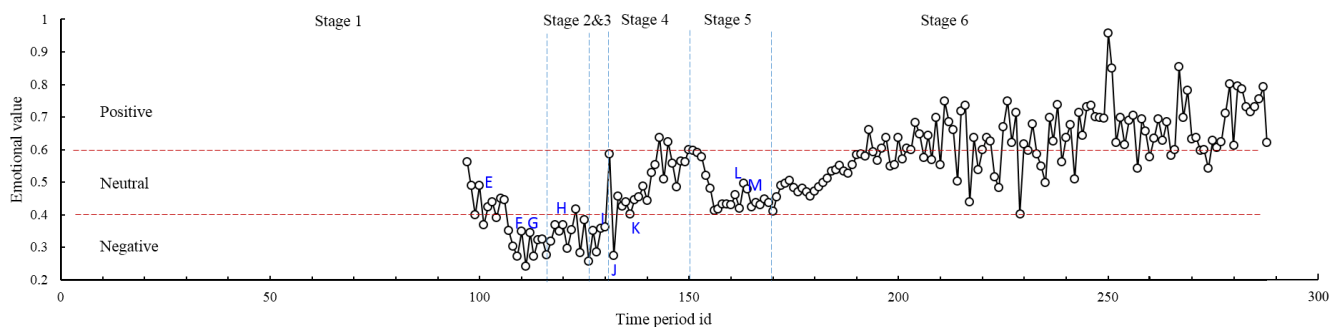


Fig. 12 Emotional evolution of social media data in Zhengzhou Metro flood.

the emotional value gradually increased, indicating that the timely release of rescue results can help guide public opinion toward positive development.

3.4 Evolution characteristics of hot topics in social media data

Understanding the main content of social media texts, especially meaningful keywords, can provide auxiliary guidance for emergency response. However, when

there are too many keywords, it becomes challenging to grasp the content of social media texts promptly. Therefore, it is crucial to classify a large number of keywords. In this section, this study categorizes all keywords into 6 topics and discusses them separately in the 6 stages of the Zhengzhou Metro flood, similar to the previous analysis.

In order to analyze the evolution characteristics of hot topics in social media data, the word frequency analysis

of social media texts is conducted. According to the analysis method in Section 2.3, this study conducts Chinese word segmentation for the social media texts in the Zhengzhou Metro flood. After stop word filtering, the TF-IDF of each keyword is calculated. The results and sorting of TF-IDF are shown in Table 6.

In the social media texts of Zhengzhou Metro flood, TF-IDF sorting and frequency sorting are basically the same. Some keywords with lower frequency have higher TF-IDF, such as “forced to stop” and “measure”. Although the frequency of these words is small, they are keywords in the accident. It shows that the TF-IDF method can discover keywords with low frequency. Among all the keywords, the word with the highest TF-IDF is “rainstorm”, which is the basic description of DE. The word with the second highest TF-IDF is “trapped”, which describes the impact of disasters on passengers. According to the analysis method in Section 2.3, the keywords with higher TF-IDF are classified, as shown in Table 7. Each classification of keywords was taken as a whole, and its proportion in social media data is calculated, as shown in Fig. 13.

In different stages of the emergency, social media

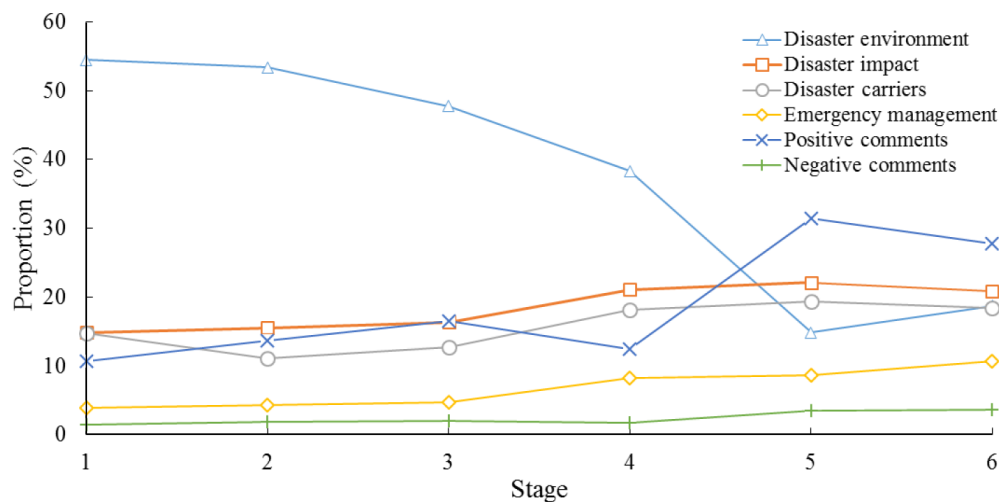
focuses on different topics. From Stage 1 to Stage 3, the main content of social media is mostly the description of disaster environment, and the proportion of DE is 54.4%, 53.4%, and 47.8%, respectively. The proportion of DI, DC, and PC is between 10% to 20%. The proportion of EM and NC is less than 5%. The proportion of hot topics in stages 1–3 is basically stable, which can represent the main distribution of social media data under non-emergency conditions. From Stage 4 to Stage 6, the distribution of hot topics of social media data changes significantly. While waiting for rescue (Stage 4), the focus of some social media evolves from DE to DI, DC, and EM. At the same time, the proportion of PC decreases from 16.5% to 12.5%. In Stage 5, with the increase of emergency rescue information, the popularity of social media in DE decreases to 14.8%. In all social media data in this stage, more people express their views. The proportion of PC is 31.5%, and the proportion of NC is 3.6%. It is worth noting that from Stage 5 to Stage 6, although the proportion of PC is the largest, the ratio of PC to NC decreases. In the 6 stages, the quantity ratios of PC to NC are 7.62, 7.36, 8.36, 7.40, 8.84, and 7.59. Therefore, in the public opinion

Table 6 Results of word frequency analysis in social media texts

Sort	Keyword	TF-IDF	Total frequency	Sort	Keyword	TF-IDF	Total frequency
1	Rainstorm	0.0152	10,164	26	Mutual aid	0.0039	1002
2	Trapped	0.0132	8432	27	Interval	0.0038	873
3	Carriage	0.0109	5413	28	Metro exit	0.0036	772
4	Hope	0.0098	4654	29	Urgently	0.0036	807
5	Rescue	0.0096	3627	30	Continuous	0.0035	824
6	Safe	0.0094	8151	31	Fearful	0.0035	859
7	Waterfall	0.0071	2551	32	Situation	0.0034	828
8	Ponding	0.0069	1806	33	Flood	0.0034	805
9	Passenger	0.0069	1831	34	City	0.0034	769
10	Encourage	0.0063	1892	35	Water level	0.0034	774
11	Safety	0.0062	2060	36	Whole network	0.0034	791
12	Forced to stop	0.0059	825	37	Caution	0.0033	788
13	Impact	0.0057	1330	38	Evacuation	0.0033	713
14	Personnel	0.0049	1533	39	Worried	0.0033	796
15	Suspend	0.0048	1450	40	Bless	0.0032	688
16	Hold on	0.0048	1335	41	Bus	0.0031	688
17	Measure	0.0046	691	42	Road	0.0031	567
18	Fire control	0.0046	1112	43	Resident	0.0031	634
19	Flooded	0.0044	1165	44	Rainfall	0.0030	879
20	Return home	0.0043	1099	45	Suffocate	0.0030	712
21	Airport	0.0041	665	46	Community	0.0030	661
22	Black-out	0.0041	1047	47	Volunteer	0.0029	507
23	Heavy rain	0.0040	967	48	Train	0.0028	578
24	Heavy rainfall	0.0040	1026	49	Slippery	0.0027	293
25	Raining	0.0039	999	50	Flood control	0.0027	498

Table 7 Classification of social media keywords in Zhengzhou Metro flood

Number	Classification	Description	Keyword
Class 1	DE	Objective description of DE and background of emergencies	Rainstorm, waterfall, ponding, heavy rain, heavy rainfall, raining, continuous, situation, flood, water level, rainfall, slippery
Class 2	DI	Impact of emergencies on metro system and passengers	Trapped, forced to stop, impact, suspend, flooded, black-out, suffocate, return home
Class 3	EM	Emergency response of the metro operation company and external rescue forces	Rescue, measure, fire control, mutual aid, urgently, evacuation, volunteer, flood control
Class 4	DC	DCs affected by emergencies, including personnel, equipment, and urban transportation system	Passenger, carriage, personnel, airport, interval, metro exit, city, whole network, bus, road, resident, community, train
Class 5	PC	Positive views expressed by the public, including encouragement, blessing, etc.	Hope, safe, encourage, safety, hold on, caution, bless
Class 6	NC	Negative views expressed by the public, including fear, sadness, etc.	Fearful, worried

**Fig. 13** Popularity change curves of social media hot topics in Zhengzhou Metro flood.

management of metro emergencies, not only the proportion of PCs, but also the ratio of positive to NCs should be concerned. Appropriate measures should be taken to guide the public's views towards positive and reduce the proportion of NCs.

4 Discussion

This study proposes a method to analyze the evolution characteristics of social media data in metro emergencies. Zhengzhou Metro flood is taken as an example to analyze the evolution characteristics of social media data, and the shortcomings in emergency response to Zhengzhou Metro flood are found. The proposed method can be applied to the analysis of social media data and provide support for metro operation in other cities (including coastal cities and inland cities). It can provide suggestions for publicity and public opinion guidance, so as to appease people's feelings and guide the people to fight against disasters in a unified way. The above results reveal that the disaster warning and public opinion guidance of Zhengzhou Metro still need to be improved. Therefore,

some suggestions for metro operation in inland cities during emergencies can be put forward through the proposed method. The specific suggestions are as follows.

(1) The metro operators in inland cities should attach great importance to the impact of the rainstorm and take timely early warning measures for the rainstorm. The metro operation company should closely cooperate with China's meteorological authority. When China's meteorological authority sends out rainstorm warnings, preventive measures in the metro system should be taken simultaneously. At the same time, early warning notices should be published by social media to remind passengers to reduce unnecessary travel, and the number of passengers entering metro stations should be controlled.

(2) The media publicity and public opinion guidance in metro emergencies should be strengthened. In each stage of emergencies, the zero reporting system should be implemented in media publicity. In the face of massive social media data, the public's concerns should be mined and responded in time, so as to reduce the public's doubts about emergency response capacity.

(3) The metro operators should pay more attention to the distress messages published by the public and respond timely. Firstly, distress messages published by the public should be monitored in real time. Secondly, the stations with high social media popularity should be mined by spatial distribution analysis. Then, the stations with severe accidents or temporary shelters around them should be further distinguished. After that, timely measures should be taken to support the affected stations and guide the people in need to the surrounding temporary shelters. Meanwhile, the rescue situation should be announced by the official media to reduce public concern.

5 Conclusions

In this study, a method to analyze the evolutionary characteristics of social media data is proposed, which is applicable to the safe operation and risk management of metros in case of emergencies. Zhengzhou Metro flood is taken as an example to discuss the evolutionary characteristics of spatio-temporal distribution, emotional distribution, and hot topics. The proposed methods and acquired results can provide support for social media data analysis and public opinion guidance of metro emergencies in other cities. The main conclusions are as follows.

(1) The time distribution of social media data in metro emergencies is greatly affected by official media, and social media popularity reached its peak after the start of emergency rescue and the completion of evacuation. The spatial distribution of social media data in metro emergencies reflects the distribution of severely affected stations and temporary shelters, which helps to take emergency rescue measures and guide passengers to take refuge.

(2) The change in people's emotions is affected by the severity of emergencies and the official media. As the severity of the disaster increases, people's emotional value decreases. With the positive report from the official media, the public's emotional value gradually increases. However, improper official media publishing will lead to a rapid decline in the emotional value of social media data and increase public doubts about emergency response capacity. Timely reports of the rescue results can help guide people's emotions to become positive.

(3) Combined with the public security triangular framework, a classification method of hot topics in metro emergencies is proposed. Hot topics are divided into 6 classifications, including DE, DI, DC, EM, PC, and NC. Before the emergency rescue, the proportion of DE is about 50%, and the proportion of DI, DC, and PC is 10%–20%. With the advance of emergency rescue, the proportion of DE decreases, and the proportion of social media data of other classifications increases. While the proportion of PC increases, the proportion of NC may

also increase, which may results in a decrease in the ratio of PC to NC and requires attention and timely guidance.

Analyzing the evolutionary characteristics of social media data in metro emergencies can provide support for the safe operation and public opinion guidance of metros. In future research, the scope of social media data used is expected to expand, with the introduction of data from platforms such as WeChat. Besides, it is expected to further explore more specific disaster relief needs from social media texts. The stations with severe accidents or temporary shelters around them can be further distinguished on the basis of this study. If the specific needs of the public can be obtained from social media texts, such as medicine, clothing, diet, daily necessities, etc., it can help emergency decision-makers to allocate emergency supplies. Through real-time monitoring of social media data and the spatio-temporal distribution of people's needs in emergencies, more accurate response and rescue measures can be implemented and dynamically adjusted.

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Declaration of competing interest

The authors have no competing interests to declare that are relevant to the content of this article.

Author contribution statement

All authors have given approval to the final version of the manuscript.

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